Problem Set for 08 May 2025: A New Jordan form over $\mathbb{C}$

> **Important**
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> Throughout, $\mathbb{C}$ is the base field. Define $|z| := \sqrt{z\overline{z}}$ for $z \in \mathbb{C}$.

Definition (Ceigen-). Let $A \in \mathbb{C}^{n\times n}$. We say that (v, λ) is a ceigenpair of A provided $A\overline{v} = \lambda v$ for some $\lambda \in \mathbb{C}$. The vector v is referred to as a ceigenvector and λ a ceigenvalue, correspondingly. For matrices, we say A is csimilar to B provided $A = PB\overline{P}^{-1}$ for some non-singular matrix P.

> **Note**
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> You can firstly try Exercise 13-15.

Exercise 1 Characterise the ceigenspace corresponding to the ceigenvalue 0.

> **Warning**
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> Ceigenspaces associated with ceigenvalues $\neq 0$ are not, in general, $\mathbb{C}$-vector spaces.

Exercise 2 Demonstrate that if λ is a ceigenvalue, then so too is $e^{i\theta} \cdot \lambda$. Furthermore, show that for any ceigenvector v, there exists a ceigenvector associated with a ceigenvalue ≥ 0.

Exercise 3 Suppose (v, σ) is a ceigenpair of A; then $(v, \sigma\overline{\sigma})$ is an eigenpair of $A\overline{A}$. In particular,

- if $A \in \mathbb{C}^{d\times d}$ possesses d ceigenpairs which are linearly independent over $\mathbb{C}$, then $A\overline{A}$ is diagonalisable with all eigenvalues non-negative.

Now consider the following **partial converse statement**:

- A has no ceigenvectors if and only if $A\overline{A}$ admits no eigenvalues in $\mathbb{R}_{\geq 0}$;
- $A \in \mathbb{C}^{d\times d}$ has d linearly independent ceigenpairs over $\mathbb{C}$ if and only if $A\overline{A}$ is diagonalisable with non-negative eigenvalues.

Prove the **partial converse statement**: suppose that $(v, \sigma\overline{\sigma})$ is an eigenpair of $A\overline{A}$; then σ is a ceigenvalue of A.

> **Tip**
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> Construct an associated ceigenvector of some $e^{i\theta}\sigma$, then invoke previous exercises.

> **Warning**
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> When A has geometric multiplicity 1, one may simply take the ceigenvector $e^{i\theta}v$; however, in general, $e^{i\theta}v$ is not a ceigenvector. The construction closely resembles that found in **Exercise 6**.

Exercise 4 Elucidate **Exercise 3** (particularly the warning) with the following example:

$$A = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

Exercise 5 This serves to complete **Exercise 2**. Suppose $\{(v_i, \lambda_i)\}_{i=1}^n$ are ceigenpairs of A such that $|\lambda_i| \neq |\lambda_j|$; show that $\{v_i\}_{i=1}^n$ are linearly independent.

Exercise 6 We aim to identify the analogue of the Jordan canonical form in the c-version, namely, $X = PJ\overline{P}^{-1}$. We commence with a very special case:

- $P\overline{P} = I$ if and only if $P = Q\overline{Q}^{-1}$ for some invertible Q.

> **Tip**
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> Take $Q = a\overline{P} + bI$.

Exercise 7 The most elementary Jordan matrix is the diagonal matrix. Show that A is cdiagonalisable (i.e., $A = PD\overline{P}^{-1}$ for some P) if and only if $A\overline{A}$ is diagonalisable and $\operatorname{rank}(A) = \operatorname{rank}(A\overline{A})$. In this scenario, $A\overline{A}$ has only non-negative eigenvalues.

> **Note**
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> In this diagonalisable case, one may also verify that $\operatorname{rank}(A) = \operatorname{rank}(A\overline{A})$; that is, the multiplicity of 0 is preserved. For $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $A\overline{A} = O$ is diagonalisable yet A is not cdiagonalisable.

Exercise 8 (Nilpotent part). Show that there exists a matrix P such that $PA\overline{P}^{-1} = \left(\begin{smallmatrix} N & O \\ O & B \end{smallmatrix}\right)$, where N comprises nilpotent Jordan blocks and B is invertible.

> **Tip**
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> Consider the method we used last term to find $J_{s_i}(0)$'s. Useful hints can be found in **Exercise 11**.

Exercise 9 To analyse the Jordan form of A, we begin by studying the Jordan form of $A\overline{A}$. Demonstrate the following:

1. $A\overline{A}$ has a characteristic polynomial with real coefficients;
2. non-real Jordan blocks occur in conjugate pairs;
3. if $\dim\ker(\lambda I - A\overline{A}) = 1$ and $\lambda \in \mathbb{R}$, then $\lambda \geq 0$;
4. **(optional)** Generalise item 3 by showing that for any eigenvalue $\lambda < 0$, the dimension of the null space $\dim\ker(\lambda I - A\overline{A})^p$ is always even ($\forall p \geq 1$), implying that negative Jordan blocks occur in pairs.

> **Warning**
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> This is a straightforward exercise. Nonetheless, note that $A\overline{A}$ may possess negative eigenvalues.

Exercise 10 Show that there exists a matrix P such that $PA\overline{P}^{-1}$ is block diagonal with the following summands:

1. (Jordan type): $J_s(\lambda)$ where $\lambda \geq 0$;
2. (Half-Jordan type): Let $H_{2s}(\lambda) := \left(\begin{smallmatrix} O & J_s(\lambda) \\ I & O \end{smallmatrix}\right)$.
    1. (For non-real eigenvalues): Since each complex Jordan block in $A\overline{A}$ appears in a pair, i.e., $\left(\begin{smallmatrix} J_s(z) & O \\ O & J_s(\overline{z}) \end{smallmatrix}\right)$, define $H_{2s}(z)$ as a block of $PA\overline{P}^{-1}$ satisfying
    $$H_{2s}(z) = \begin{pmatrix} O & I \\ J_s(z) & O \end{pmatrix}.$$
    2. (For negative eigenvalues): For a negative eigenvalue λ of A, the Jordan blocks also appear in pairs (see **Exercise 9**).

We refer to this as the cJordan form of A. It is uniquely determined by A.

> **Tip**
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> The uniqueness (as we equal $H(z)$ and $H(\overline{z})$) of non-zero J- and H-blocks is governed by $A\overline{A}$; the uniqueness of the $J(0)$-portion is determined by the rank sequence $\operatorname{rank}(A)$, $\operatorname{rank}(A\overline{A})$, $\operatorname{rank}(A\overline{A}A)$, and so forth.

Exercise 11 (optional) (The uniqueness between the cJordan form of A and the Jordan form of $A\overline{A}$). The cJordan form is essentially unique.

- Find distinct matrices $A \neq B$ such that $A\overline{A} = B\overline{B}$;

- Suppose $A\overline{A} = B\overline{B}$ are invertible; then there exists a matrix P such that $PA\overline{P}^{-1} = B$. In this case, we say A is scimilar to B.

In summary, A is csimilar to B if and only if the following hold:

1. $A\overline{A}$ is similar to $B\overline{B}$;
2. $\operatorname{rank}(A\overline{A}A\cdots A(\overline{A})) \equiv \operatorname{rank}(B\overline{B}B\cdots B(\overline{B}))$; that is, A and B possess the same $J(0)$-portion in their cJordan forms.

Exercise 12 (optional) Utilise the cJordan form to show that any complex matrix A is csimilar to:

1. λA for $|\lambda| = 1$;
2. $\overline{A}$;
3. A^T;
4. a real matrix;
5. a Hermitian matrix.

Exercise 13 (optional) Show that for any complex square matrix A,

$$\det(AA^H + I) \geq \det(A\overline{A} + I) > 0.$$

> **Tip**
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> Apply the identity $AA^T + BB^T = RR^T$, and the Cauchy–Binet formula to establish the first inequality.

Exercise 14 (optional) Using cJordan forms, show that any complex matrix is the product of a symmetric matrix and a Hermitian matrix.

> **Note**
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> It is known that any real or complex matrix can be expressed as the product of two symmetric matrices (see exercises from the past semesters).

Exercise 15 (optional) Show that $\begin{pmatrix} O & A \\ \overline{A} & O \end{pmatrix}$ is similar to $\begin{pmatrix} O & B \\ \overline{B} & O \end{pmatrix}$, if and only if $\begin{pmatrix} O & A \\ -\overline{A} & O \end{pmatrix}$ is similar to $\begin{pmatrix} O & B \\ -\overline{B} & O \end{pmatrix}$.

> **Tip**
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> Equivalently, if and only if A and B are csimilar.